TESTs - Lesson 09 - MyERC20.sol and MyERC721.sol

Who is responsible for proposing and guiding changes within the Ethereum ecosystem through EIPs?

A Vitalik Buterin and the Ethereum Foundation

B Node operators and smart contract developers

C A centralized governing body

What is an 'Idea' status in the EIPs status terms?

A A formally tracked stage of an EIP in development

B An EIP that has been implemented and is now a standard

C An informally tracked pre-draft that is not yet in the EIP repository

Which of the following ERC-20 functions is mandatory for returning the total token supply?

A name()

B symbol()

C totalSupply()

D decimals()

What is the primary benefit of using the OpenZeppelin Contracts library for smart contract development?

A It provides a new programming language for smart contracts.

B It offers reusable, community-vetted Solidity components for secure development.

C It automatically deploys contracts to the Ethereum mainnet.

What is the primary function of Role-Based Access Control (RBAC) in smart contracts, as implemented by OpenZeppelin?

A To allow only a single, designated owner to manage the contract.

B To define different levels of authorization and permissions for various addresses or groups.

C To restrict all contract functions to be private.

What is the significance of using 'decimals' in an ERC-20 token contract?

A It defines the token's total supply capacity.

B It specifies the number of decimal places for user representation of the token value.

C It determines the token's fungibility.

What is the expected behavior if an account without the MINTER_ROLE attempts to mint new tokens in an OpenZeppelin ERC-20 contract configured with RBAC?

A The transaction will succeed, but the tokens will be burned immediately.

B The transaction will revert with an 'AccessControlUnauthorizedAccount' error.

C The transaction will automatically assign the MINTER_ROLE to the account before minting.

What are Solidity 'events' primarily used for in dApps?

A To store large amounts of data directly on the blockchain efficiently.

B To create logs that can be listened to by off-chain applications to react to contract changes.

C To enable direct interaction between different smart contracts without external calls.

Which of the following best describes the principle of 'inheritance' in Solidity, as demonstrated with OpenZeppelin contracts?

A It allows a contract to store its entire state on an external database.

B It enables a contract to copy and extend functionalities (state variables, functions) from other contracts.

C It ensures that a contract's code can never be modified after deployment.